ECE PROJECT · EMBEDDED SYSTEMS

# AuraReflex

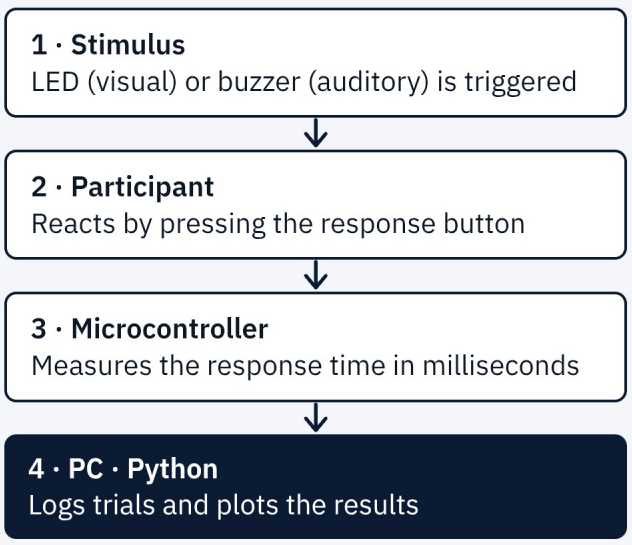
Real-Time Visual (LED) & Auditory (Buzzer) Reaction Time Analysis


## Abstract

AuraReflex is a real-time reaction-time measurement system built for our ECE project. It presents either a visual stimulus (LED) or an auditory stimulus (buzzer), records how long a participant takes to respond, and plots the results for analysis.

Two participants each completed three trials per stimulus. Auditory responses were consistently faster: average reaction times were 782.71 ms and 1248.18 ms for the buzzer, against 3698.36 ms and 4765.75 ms for the LED. The fastest single response was 270.17 ms (buzzer), and both participants got faster from trial to trial.


## Figure



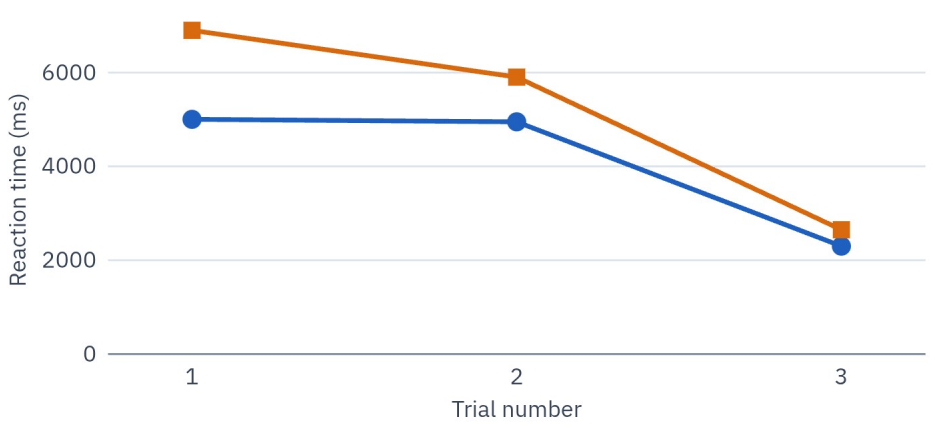

[ADD PHOTO OF YOUR HARDWARE SETUP]

Figure 1. AuraReflex system block diagram

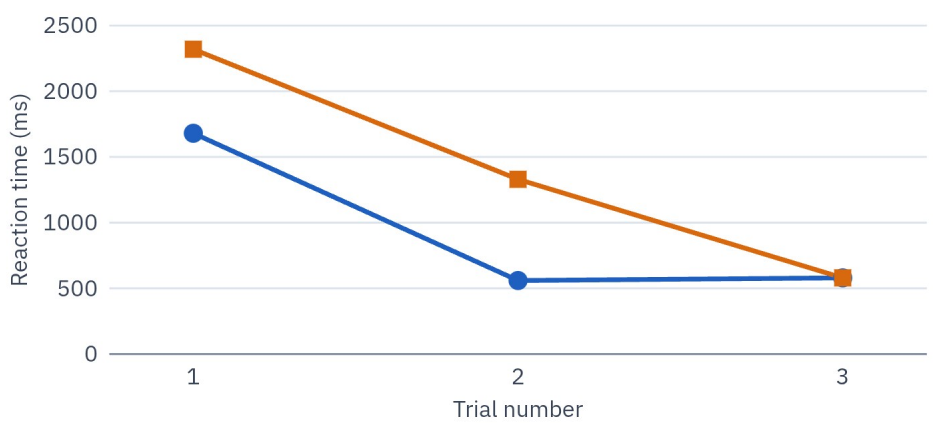
## Simulations & Graphs

Participant 1 Participant 2

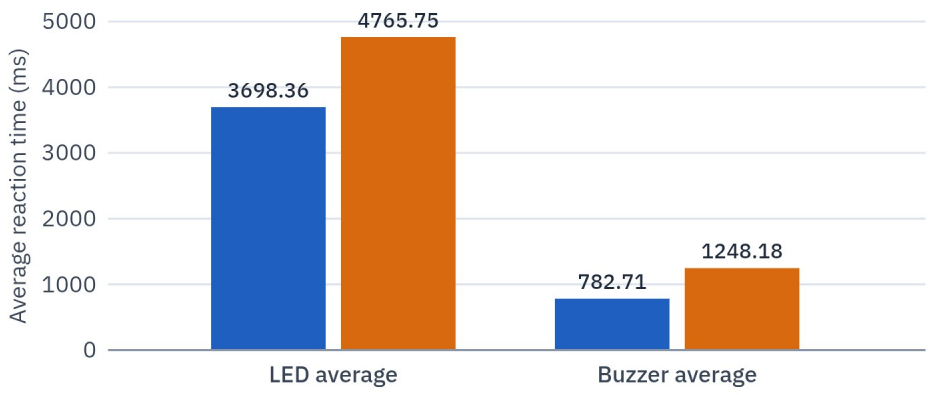
### 1. LED test (visual): 3-trial reaction curve



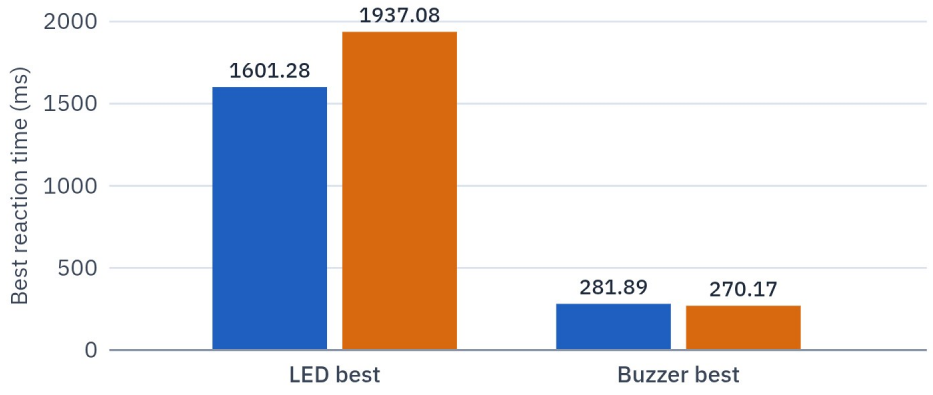

### 2. Buzzer test (auditory): 3-trial reaction curve




### 3. Combined average response time comparison




### 4. Combined best (fastest) response time comparison




**270.17 ms** fastest response recorded (Participant 2, buzzer)

**3.8–4.7×** faster average response to sound than to light

**12 trials** 2 participants × 2 stimuli × 3 trials each

## Applications

- **Driver alertness checks** – screen reaction speed before long drives.
- **Sports & reflex training** – track an athlete's progress trial by trial.
- **Cognitive screening** – a low-cost aid for baseline response monitoring.
- **Gaming & e-sports** – benchmark visual versus audio response.
- **Industrial safety & HMI design** – pick alarm cues (light or sound) that operators react to fastest.

## References


1. Woods, D. L., et al. (2015). Factors influencing the latency of simple reaction time. *Frontiers in Human Neuroscience*, 9, 131.
2. Shelton, J., & Kumar, G. P. (2010). Comparison between auditory and visual simple reaction times. *Neuroscience & Medicine*, 1(1), 30–32.
3. Kosinski, R. J. (2008). *A Literature Review on Reaction Time.* Clemson University.
4. Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95.
5. Project source code (GitHub): [ADD YOUR REPOSITORY LINK]



GROUP MEMBERS

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